

Glossary

Advanced Data Analytics



Terms and definitions from Course 4

A

Absolute values: (Refer to **observed values**)

Accuracy: The proportion of data points that were correctly categorized

Adjusted R^2 : A variation of R^2 that accounts for having multiple independent variables present in a linear regression model

Analysis of Variance (ANOVA): A group of statistical techniques that test the difference of means between three or more groups

ANCOVA (Analysis of Covariance): A statistical technique that tests the difference of means between three or more groups while controlling for the effects of covariates, or variable(s) irrelevant to the test

B

Backward elimination: A stepwise variable selection process that begins with the full model, with all possible independent variables, and removes the independent variable that adds the least explanatory power to the model

Best fit line: The line that fits the data best by minimizing some loss function or error

Bias: Simplifying the model predictions by making assumptions about the variable relationships

Bias-variance trade-off: Balance between two model qualities, bias and variance, to minimize overall error for unobserved data

Binomial logistic regression: A technique that models the probability of an observation falling into one of two categories, based on one or more independent variables

Binomial logistic regression linearity assumption: An assumption stating that there should be a linear relationship between each X variable and the logit of the probability that Y equals one

C

Causation: Describes a cause-and-effect relationship where one variable directly causes the other to change in a particular way

Chi-squared (χ^2) Goodness of Fit Test: A hypothesis test that determines whether an observed categorical variable follows an expected distribution

Chi-squared (χ^2) Test for Independence: A hypothesis test that determines whether or not two categorical variables are associated with each other

Confidence band: The area surrounding a line that describes the uncertainty around the predicted outcome at every value of X

Confusion matrix: A graphical representation of how accurate a classifier is at predicting the labels for a categorical variable

Confidence interval: A range of values that describes the uncertainty surrounding an estimate

Correlation: Measures the way two variables tend to change together

D

Dependent variable (Y): The variable a given model estimates

E

Errors: In a regression model, the natural noise assumed to be in a model

Explanatory variable: (Refer to **independent variable**)

Extra Sum of Squares F-test: A test that quantifies the difference between the amount of variance that is left unexplained by a reduced model that is explained by the full model

F

Feature selection: (Refer to **variable selection**)

Forward selection: A stepwise variable selection process that begins with the null model—with 0 independent variables—considers all possible variables to add; incorporates the independent variable that contributes the most explanatory power to the model

H

Hold-out sample: A random sample of observed data that is not used to fit the model

Homoscedasticity assumption: An assumption of simple linear regression stating that the variation of the residuals (errors) is constant or similar across the model

Hypothesis testing: A statistical procedure that uses sample data to evaluate an assumption about a population parameter

I

Independent observation assumption: An assumption of simple linear regression stating that each observation in the dataset is independent

Independent variable (X): The variable whose trends are associated with the dependent variable

Interaction term: The term that represents how the relationship between two independent variables is associated with changes in the mean of the dependent variable

Intercept (constant B_0): The y value of the point on the regression line where it intersects with the y-axis

L

Likelihood: The probability of observing the actual data, given some set of beta parameters

Line: A collection of an infinite number of points extending in two opposite directions

Linearity assumption: An assumption of simple linear regression stating that each predictor variable (X_i) is linearly related to the outcome variable (Y)

Linear regression: A technique that estimates the linear relationship between a continuous dependent variable and one or more independent variables

Link function: A nonlinear function that connects or links the dependent variable to the independent variables mathematically

Logistic regression: A technique that models a categorical dependent variable (Y) based on one or more independent variables (X)

Loss function: A function that measures the distance between the observed values and the model's estimated values

Logit: The logarithm of the odds of a given probability

Log-Odds function: (Refer to **logit**)

M

MAE (Mean Absolute Error): The average of the absolute difference between the predicted and actual values

MANCOVA (Multivariate Analysis of Covariance): An extension of ANCOVA and MANOVA that compares how two or more continuous outcome variables vary according to categorical independent variables, while controlling for covariates

MANOVA (Multivariate Analysis of Variance): An extension of ANOVA that compares how two or more continuous outcome variables vary according to categorical independent variables

Maximum Likelihood Estimation (MLE): A technique for estimating the beta parameters that maximize the likelihood of the model producing the observed data

Model assumptions: Statements about the data that must be true in order to justify the use of a particular modeling technique

MSE (Mean Squared Error): The average of the squared difference between the predicted and actual values

Multiple linear regression: A technique that estimates the relationship between one continuous dependent variable and two or more independent variables

Multiple regression: (Refer to **multiple linear regression**)

N

Negative correlation: An inverse relationship between two variables, where when one variable increases, the other variable tends to decrease, and vice versa

Normality assumption: An assumption of simple linear regression stating that the residuals are normally distributed

No multicollinearity assumption: An assumption of simple linear regression stating that no two independent variables (X_i and X_j) can be highly correlated with each other

O

Observed values: The existing sample of data, where each data point in the sample is represented by an observed value of the dependent variable and an observed value of the independent variable

One hot encoding: A data transformation technique that turns one categorical variable into several binary variables

One-Way ANOVA: A type of statistical testing that compares the means of one continuous dependent variable based on three or more groups of one categorical variable

Ordinary least squares estimation (OLS): A common way to calculate linear regression coefficients

Outcome variable (Y): (Refer to **dependent variable**)

Overfitting: When a model fits the observed or training data too specifically and is unable to generate suitable estimates for the general population

P

P-value: The probability of observing results as extreme as those observed when the null hypothesis is true

Positive correlation: A relationship between two variables that tend to increase or decrease together

Post hoc test: An ANOVA test that performs a pairwise comparison between all available groups while controlling for the error rate

Precision: The proportion of positive predictions that were true positives

Predicted values: The estimated Y values for each X calculated by a model

Predictor variable: (Refer to **independent variable**)

R

R^2 (The Coefficient of Determination): The coefficient that measures the proportion of variation in the dependent variable, Y, explained by the independent variable(s), X

Recall: The proportion of positives the model was able to identify correctly

Regression analysis: A group of statistical techniques that use existing data to estimate the relationships between a single dependent variable and one or more independent variables

Regression coefficient: The estimated betas in a regression model

Regression models: (Refer to **regression analysis**)

Regularization: A set of regression techniques that shrinks regression coefficient estimates towards zero, adding in bias, to reduce variance

Residual: The difference between observed or actual values and the predicted values of the regression line

Response variable: (Refer to **dependent variable**)

S

Scatterplot matrix: A series of scatterplots that show the relationships between pairs of variables

Simple linear regression: A technique that estimates the linear relationship between one independent variable, X, and one continuous dependent variable, Y

Slope: The amount that y increases or decreases per one-unit increase of x

Sum of squared residuals (SSR): The sum of the squared difference between each observed value and its associated predicted value

T

Two-Way ANOVA: A type of statistical testing that compares the means of one continuous dependent variable based on three or more groups of two categorical variables

V

Variable selection: The process of determining which variables or features to include in a given model

Variance: Model flexibility and complexity, so the model learns from existing data

Variance inflation factors (VIF): Factors that quantify how correlated each independent variable is with all of the other independent variables